



Final exam - ECE 124

Term: Spring Year: 2021

Student name: _____
(First name) (Middle name) (Last name)

Waterloo student identification number _____

Course abbreviation and number: **ECE 124**

Course title: Digital Circuits and Systems

Name of instructor(s): Anwar Hasan

Exam start time: 11:00am (Eastern), Saturday, August 14, 2021

Exam end time: 11:00am (Eastern), Sunday, August 15, 2021

Duration: 24 hours

Exam type: Open-book and take home

Other instructions:

- As you write the exam, you must maintain academic integrity. This is important for learning.
- For this exam, you must not communicate with or seek help from anyone.
- You must also not share this exam or any part of it in any form with anyone who has not yet submitted this exam or a replacement of it for grading.
- By submitting the exam for grading, you confirm that you have followed the above rules and committed no academic offences such as cheating and plagiarism.
- Penalty for late submission – 1% per minute.
- Answer all questions in this exam. Clearly state any assumptions you make when answering any questions.
- Show your work—no magic answers are acceptable. Use additional pages as needed.
- For each question, upload on Crowdmark your answer to that question only, ensuring that the uploaded file does not contain answers to any other questions. Otherwise, your answers risk not being marked or 10% penalty. There is no need to upload this instruction page on Crowdmark.
- If you believe the exam has errors/mistakes, please bring them to the instructor's attention by sending an email to ahasan@uwaterloo.ca. However, do not contact the instructor or any TAs to seek clarification of any questions.

1. [10 marks]

Suppose a coin-operated vending machine dispenses face masks under the following conditions:

- The machine accepts loonies and toonies, but only one coin at a time
- It takes \$3.00 for one pack of face masks to be dispensed from the machine
- If the deposited amount is \$4.00, the machine dispenses one pack of face masks and returns \$1.00
- The machine's inputs are disabled while it is giving out face masks and, if applicable, any changes.

The machine has no "Cancel" button.

The control of the vending machine is to be designed using an **asynchronous** sequential circuit. To indicate the receipt of a coin, the circuit has two input signals L and T. Signal L (respectively T) is 1 when the machine senses the coin being inserted is a loonie (respectively a toonie); otherwise both L and T are 0. The circuit operation is much faster than the coin insertion process, i.e., both L and T are 0 during the time gap between two consecutive insertions.

The circuit has two output signals z_1 and z_2 . Signal z_1 is 1 to dispense a pack of face masks, otherwise z_1 is 0. Signal z_2 is 1 to return a loonie, otherwise z_2 is 0.

Draw a state diagram for the control of the vending machine assuming the **Mealy** model. Give justification of your diagram. Also give the corresponding flow table. [Draw your state diagram neatly.]

2. [7 marks]

Denote N as the number that represents the rightmost two digits of your UW student ID (e.g., $N = 56$ if your ID is 20123456). Divide your N by 5 and denote the resulting remainder as R . Now let $V = R + 9$ and denote the least significant four bits of the binary representation of V as $(v_3 v_2 v_1 v_0)$, where $v_i \in \{0,1\}$ for $0 \leq i \leq 3$. [You will use this value in this and some other questions of the exam.]

In the flow table below, replace each v_i ($0 \leq i \leq 3$) with its value you determined above. Then identify *compatible* states that may exist in the following flow table, draw the corresponding merger diagram and give the reduced flow table. (You do not need to apply the merging procedure multiple times.)

Present state	Next state					Output
	$w_2 w_1 =$	00	01	10	11	z
A		(A)	G	E	-	v_3
B		F	-	(B)	D	v_2
C		F	(C)	-	H	v_1
D		-	C	E	(D)	v_0
E		A	-	(E)	D	v_0
F		(F)	C	B	-	v_1
G		F	(G)	-	D	v_2
H		-	-	E	(H)	v_3

3. [6 marks]

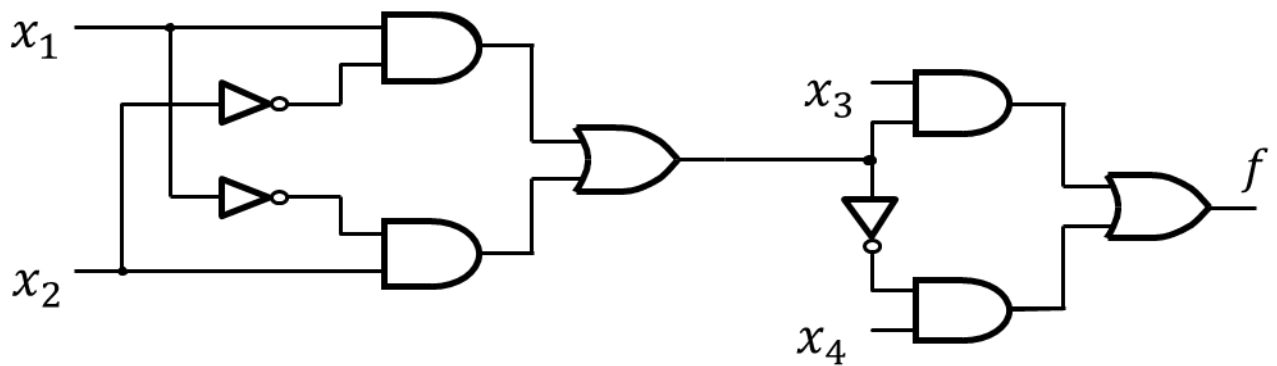
[Before attempting this question, determine v_i 's ($0 \leq i \leq 3$) in Q #2 and write them here at the beginning of your answer.]

Below is a flow table for an asynchronous FSM. Find a race-free assignment for the states by using the one-hot-encoding scheme. To start the encoding, assume that A=1000, B=0100, C=0010 and D=0001. Give the modified flow table.

Present state	Next state					Output
	$w_2w_1 =$	00	01	10	11	
A		B	A	D	A	v_2v_3
B		B	B	D	A	v_0v_1
C		C	A	C	D	v_1v_0
D		C	B	D	D	v_3v_2

4. [7 marks]

For the circuit shown below, design a corresponding circuit that is hazard free. Use only AND, OR and NOT gates. Assume that AND and OR gates can each have any number of inputs. Show your work and state your design approach.



5. [8 marks]

[Before attempting this question, determine the value of V in Q #2 and write it here at the beginning of your answer.]

A modulo- n counter produces the sequence $\{0, 1, \dots, n-1\}$. See p. 7 of slide set 6-part 5 for a modulo-6 counter, where a 3-bit up-counter is used as sub-circuit.

You are to design a circuit that would output the following sequence:

$\{0, 1, \dots, V, 1, 2, \dots, V, 2, 3, \dots, V, \dots, V-1, V\}$.

Note that the sequence can be considered to comprise V sets of numbers, namely, $(0, 1, \dots, V)$, $(1, 2, \dots, V)$, $\dots$, $(V-1, V)$, where the 1st set starts at 0, the 2nd set at 1, and so on, and each set ends at V . As an example, the sequence for $V = 3$ is given below (both in binary and decimal) and it requires nine clock cycles (t_0 to t_8).

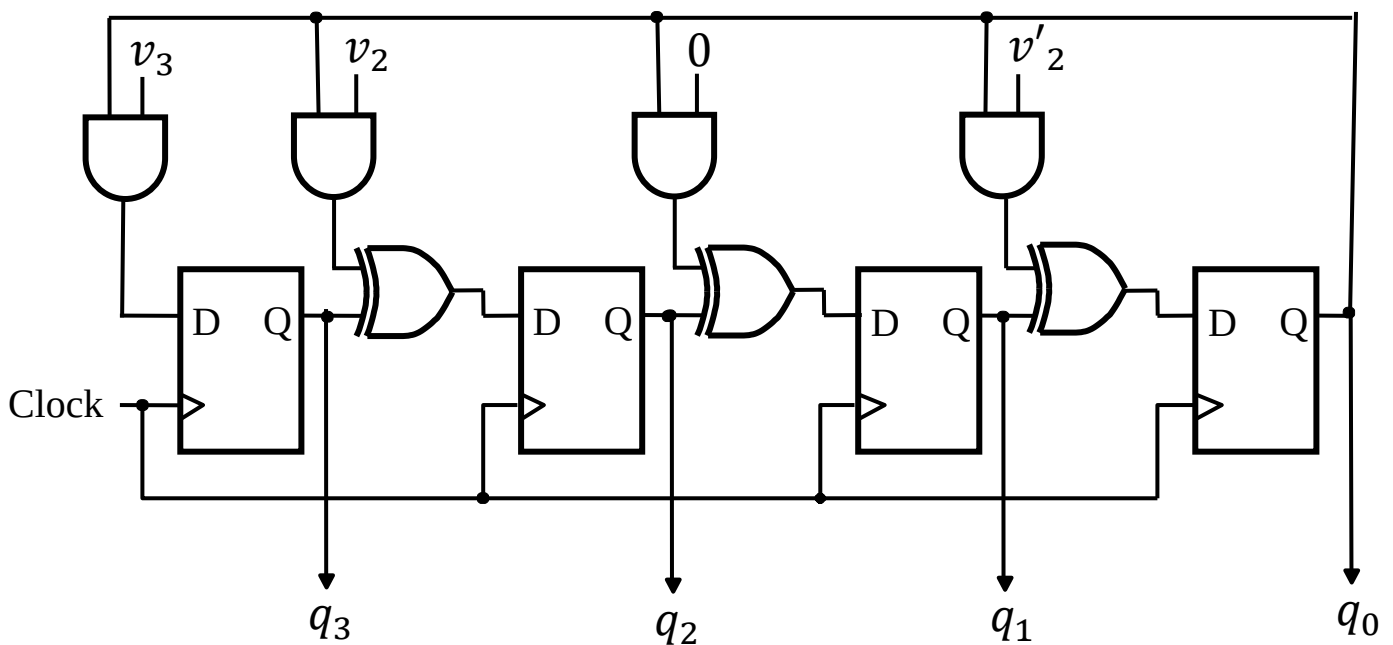
Clock cycle	Output sequence	
	binary	decimal
t_0	00	0
t_1	01	1
t_2	10	2
t_3	11	3
t_4	01	1
t_5	10	2
t_6	11	3
t_7	10	2
t_8	11	3

For your design, use no more than two 4-bit synchronous up-counters as sub-circuit, where each counter has synchronous parallel-load capability, an asynchronous Reset input for cold start, an Enable input and a Clock input. You are also allowed to use up to five multiple-input AND gates and an appropriate number of NOT gates. Draw your circuit neatly and explain how it produces the sequence. [To be considered for the full mark, give a minimum cost circuit. For this question, do not use state-diagram/table based design approaches given in slide set 7.]

6. [8 marks]

[Before attempting this question, determine v_i 's ($0 \leq i \leq 3$) in Q #2 and write them at the beginning of your answer to this question.]

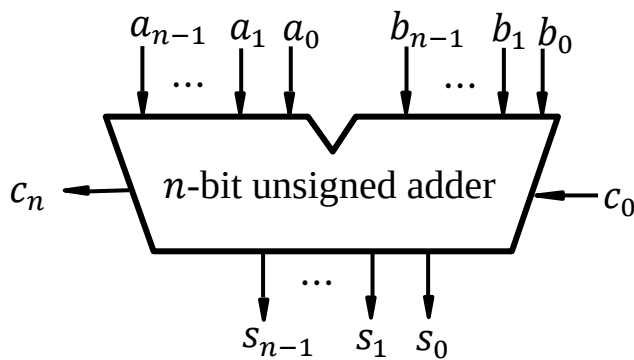
The diagram below is a shift register with feedback logic to its various stages. In the diagram, each v_i is as you derived in Q #2. The output of the shift register is labelled as $q_3q_2q_1q_0$. Assume that the shift register's output in clock cycle t_0 is 1000, which is 8 in decimal. Write the sequence the shift register will produce from this initial value (i.e., write all the output values until the register produces the initial value again). List the value of each sequence element in both binary and decimal. (See Q #5 for an example on how to list the values in a tabular format.)



7. [7 marks]

Assume that numbers are represented using n bits in the 2's complement system. You are to design a circuit that takes as input two positive integers X and Y , where $0 \leq X < 2Y$ and $0 < Y$. The circuit output is Z , where $Z = X \bmod Y$. Note that $0 \leq Z < Y$.

For your design, use n -bit unsigned adders as sub-circuit. Besides the adders, you are allowed to use a necessary number of NOT, multiple-input AND, OR and XOR gates. Draw your circuit neatly and explain how it computes $X \bmod Y$. To be considered for the full mark, give a minimum cost circuit. [Below is a block diagram of the n -bit unsigned adder where the numbers to be added are $A = (a_{n-1}, \dots, a_1, a_0)$ and $B = (b_{n-1}, \dots, b_1, b_0)$, the sum is $S = (s_{n-1}, \dots, s_1, s_0)$, and the carry-in and carry-out are c_0 and c_n , respectively.]



8. [7 marks]

The 4-bit unsigned multiplier circuit given on p. 4 of slide set 10 - part 5 requires 16 full adders (FAs) and 16 two-input AND gates. The circuit can also be used for squaring (say, A^2) by setting its both inputs (multiplicand M and multiplier Q) to A . This is however not an efficient approach as it does not take advantage of the fact that squaring requires only one number (A) as input.

Your task is to design a low cost 4-bit unsigned squaring unit. For this design, use only FAs and two-input AND gates. To be considered for the full mark, your circuit must not require more than four FAs and four 2-input AND gates. Justify your design.